Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/23/2021
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Applied Capillary Fluidics Research For Spacecraft		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mmw@cecs.pdx.edu	**Fax:**	FY 503-725-8255
PI Organization Type:	UNIVERSITY	**Phone:**	503 725-4292
Organization Name:	Portland State University		
PI Address 1:	MCECS Research		
PI Address 2:	P.O. Box 751		
PI Web Page:	https://		
City:	Portland	**State:**	OR
Zip Code:	97207-0751	**Congressional District:**	3
Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/19/2018	**End Date:**	08/20/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	4
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/20/2021 per NSSC information (Ed., 8/17/21) NOTE: End date changed to 4/10/2021 per NSSC information (Ed., 9/3/20) NOTE: End date changed to 4/10/2021 per NSSC information (Ed., 9/3/20) NOTE: End date changed to 4/10/2020 per NSSC information (Ed., 10/21/19) NOTE: End date changed to 1/18/2020 per NSSC information (Ed., 2/27/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0436		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To meet the challenges posed by NASA's plans to further explore outer space, a large body of applied science and engineering investigations must be completed to assure the development, construction, and demonstration of the highly reliable performance of myriad advanced life support systems and unit operations. The advances must occur across all subjects within the Physical Sciences discipline, but perhaps most pointedly within the Fluid Physics sub-discipline. From liquid fuels, to propellants, to cryogens, to coolants, to water processes and management, to plant and animal habitats, and others, the management of liquids without the benefit of constant gravitational separation poses significant challenges to the designers of such systems. In this work we sought to apply the results of recent low-g capillary fluidics research, performed in part aboard the International Space Station (ISS) and Portland State University Dryden Drop Tower, to deliver solutions to NASA for a variety of pressing needs. These include: Passive biofluids management, Water delivery and management for plants in space, and a variety of low-g biofluids, surface tension containment, and educational/outreach demonstrations and activities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The majority of the effort of this research centers on the development of plant watering systems for spacecraft, which have been troubling since the beginning of space flight. We are supporting modern design activities and down-selection of method(s) for water delivery to a variety of plant habitats considering hydroponics, and artificial soil for single- and multi-plant arrays. Positive research returns could mean a new era of plant growth capabilities in space. The approach takes into account low-g and partial g environments such as lunar and Martian g environments. Significant methods and design approaches and success are shared with other investigators in the NASA microgravity community.
Task Progress:	To meet the challenges posed by NASA's plans to further explore outer space, a large body of applied science and engineering investigations must be completed to assure the development, construction, and demonstration of the highly reliable performance of myriad advanced life support systems and unit operations. The advances must occur across all subjects within the Physical Sciences discipline, but perhaps most pointedly within the Fluid Physics sub-discipline. From liquid fuels, to propellants, to cryogens, to coolants, to water processes and management, to plant and animal habitats, and others, the management of liquids without the benefit of constant gravitational separation poses significant challenges to the designers of such systems. In this work we seek to apply the results of recent low-g capillary fluidics research, performed in part aboard the International Space Station (ISS) and Portland State University Dryden Drop Tower, to deliver solutions to NASA for a variety of pressing needs. These include 1. Plant Water Management (PWM) in Space (PWM, PI M. Weislogel) ; 2. Passive biofluids management (i.e., low-g Petri dishes, PI Dr. W. Nicholson, University of Florida-UF) ; 3. The Surface Tension Containment Experiment (Flight Experiment STCE, PI J. Buchli, NASA JSC) ; 4. Nickelodeon Slime in Space (ISS Flight experiment, CASIS Outreach) We note also that this work resulted in a significant number of consulting-level input to other currently funded NASA flight experiment designs. Regarding present dissemination of the work, 5 journal articles and 10 conference papers were published and/or submitted for this work as listed herein. Five MSME (Master of Science in Mechanical Engineering) students were supported in part for the work. Approximately one dozen technical presentations were also given.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Weislogel MM, Graf JC, Wollman AP, Turner CC, Cardin KJT, Torres LJ, Goodman JE, Buchli JC. "How advances in low-g plumbing enable space exploration." npjMicrogravity. 2022 May 20;8:16. https://doi.org/10.1038/s41526-022-00201-y , May-2022
Articles in Peer-reviewed Journals	McCraney J, Weislogel M, Steen P. "The draining of capillary liquids from containers with interior corners aboard the ISS." npjMicrogravity. 2021 Nov 11;7(1):45. https://doi.org/10.1038/s41526-021-00173-5 ; PMID: 34764319; PMCID: PMC8585966 , Nov-2021
Articles in Peer-reviewed Journals	Cardin K, Wang S, Desjardins O, Weislogel MM. "Rebound of large jets from superhydrophobic surfaces in low gravity." Physical Review Fluids. 2021 Jan;6(1):014003. https://doi.org/10.1103/PhysRevFluids.6.014003 , Jan-2021
Articles in Peer-reviewed Journals	Siddique A, Trimble M, Zhao F, Weislogel MM, Tan H. "Jet ejection following drop impact on micro-pillared hydrophilic substrates." Physical Review Fluids. 2020 Jun;5(6):063606. https://doi.org/10.1103/PhysRevFluids.5.063606 , Jun-2020
Articles in Peer-reviewed Journals	Avhad A, Tan H, Al-Jubaree T, Torres L, Weislogel M. "A numerical investigation of puddle jumping." Physics of Fluids. 2020 Jan;32(1):012109. https://doi.org/10.1063/1.5139235 , Jan-2020
Articles in Peer-reviewed Journals	Nicholson WL, Fajardo-Cavazos P, Turner C, Currie TM, Gregory G, Jurca T, Weislogel M. "Design and validation of a device for mitigating fluid microgravity effects in Biological Research in Canister spaceflight hardware." Front Space Technol. 22 December 2021;2:797518. https://doi.org/10.3389/frspt.2021.797518 , Dec-2021
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Articles in Peer-reviewed Journals	McCraney J, Bostwick J, Weislogel M, Steen P. "Bubble migration in containers with interior corners under microgravity conditions." Exp Fluids. 2023 Jul 27;64(8):140. https://doi.org/10.1007/s00348-023-03677-w , Jul-2023
Papers from Meeting Proceedings	Nabity J, Pitts R, Rehmeier J, MacCraiger C, Weislogel M, Escobar C, Shaffer B, Escobar A. "Capillary-driven provision of water and nutrients to plants grown in microgravity." 50th International Conference on Environmental Systems - ICES 2020, Lisbon, Portugal, July 12-16, 2020. 50th International Conference on Environmental Systems - ICES 2020, Lisbon, Portugal, July 12-16, 2020. ICES paper No. 219. , Jul-2020

Papers from Meeting Proceedings	Hatch T, Weislogel M, Mungin R, McQuillen J. "Plant Water Management in Microgravity." 50th International Conference on Environmental Systems - ICES 2020, Lisbon, Portugal, July 12-16, 2020. 50th International Conference on Environmental Systems - ICES 2020, Lisbon, Portugal, July 12-16, 2020. ICES paper No. 28. , Jul-2020
Papers from Meeting Proceedings	Weislogel M. "Giant Drops and Sprays in Low-Gravity Environments." Special Plenary Paper and Presentation. ILASS-Asia 2019, 20th Annual Conference on Liquid Atomization and Spray Systems, Ube, Japan, December 21-23, 2019. ILASS-Asia 2019, 20th Annual Conference on Liquid Atomization and Spray Systems, Special Plenary Paper and Presentation, Ube, Japan, December 21-23, 2019. 5 pages. , Dec-2019
Papers from Meeting Proceedings	Siddiuqe AU, Zhao F, Weislogel M, Tan H. "Jet Initiation after Drop Impact on Micropatterned Hydrophilic Surfaces." IMECE®2019 International Mechanical Engineering Congress & Exposition, Salt Lake City, Utah, November 11-14, 2019. Proceedings. IMECE®2019 International Mechanical Engineering Congress & Exposition, Salt Lake City, Utah, November 11-14, 2019. Paper No: IMECE2019-11500, V007T08A053. https://doi.org/10.1115/IMECE2019-11500 , Nov-2019
Papers from Meeting Proceedings	Mohler S, Weislogel M, Graf J, Soto L. "The Dynamics of Massively Parallel Open Capillary Channel Systems for Direct-Contact Liquid Sorbent Applications in Spacecraft Life Support." 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. Paper ICES-2019-235. , Jul-2019
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Papers from Meeting Proceedings	Turner C, Goodman J, Mohler S, Mungin R, Weislogel M, Ungar EK, Buchli JC. "Mitigation of Micro-Droplet Ejections During Open Cabin Unit Operations Aboard ISS." 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. Paper ICES-2019-201. , Jul-2019